



D831 Introduction to AI and Security Syllabus

Assessments: 1 Pre-Assessment, 1 Objective Assessment
Average Completion Time: 7 Weeks

Course Prerequisites

- None

Course Description

Introduction to AI and Security provides an overview of critical terminology and key concepts for artificial intelligence (AI). Learners explore the history and evolution of AI and study bias, ethical issues, and security concerns. Contextualized examples offer learners an opportunity to see these concepts in professional scenarios such as data poisoning and insider threats attacking the AI model.

Competencies

1: Describes Types of AI

The learner describes the types of artificial intelligence for decision-making in real-world applications.

2: Identifies Suitable Data Sources

The learner explains how the collection, wrangling, and cleaning of data impacts AI/ML models.

3: Explains Best Practices for Managing Secure AI Systems

The learner explains best practices for managing secure AI systems within an organization.

Competency Weighting

Here are the competency weightings for this course. The percent weight associated with each competency is an indicator of what percentage of the assessment is mapped to each competency. For example, on a 50-item (or 4-task) assessment, if the first competency is weighted 10%, then about five items on the assessment will be related

to that competency.

- 1 - 30%
- 2 - 30%
- 3 - 40%

Learning Resources

This section provides additional information on accessing the resources for this course. For many resources, WGU has provided automatic access throughout the course. You may, however, need to enroll manually or acquire resources independently. Some resources may be available in a format different from the course. Read the full instructions provided on this page to ensure you have access to all your resources in a timely manner.

- Ihezukwu, S. (2025, April 28). *Cybersecurity Awareness: AI* [Video]. LinkedIn Learning. <https://www.linkedin.com/learning/cybersecurity-awareness-ai/cybersecurity-and-ai?u=2045532 >
- Islam, R. (2025). *Generative AI, cybersecurity, and ethics*. Wiley.
- Liu, A. C.-C., Law, O. M. K., & Law, I. (2022). *Understanding artificial intelligence: Fundamentals and applications*. Wiley-IEEE Press.
- Mahajan, S., Khurana, M., & Estrela, V. V. (2024). *Applying artificial intelligence in cybersecurity analytics and cyber threat detection*. John Wiley & Sons, Inc.
- Santos, O. & Pearson (2025, January 23). *Securing Generative AI: Strategies, Methodologies, Tools, and Best Practices* [Video]. LinkedIn Learning. <https://www.linkedin.com/learning/securing-generative-ai-strategies-methodologies-tools-and-best-practices/understanding-the-risks-of-the-ai-and-ml-supply-chain?resume=false&u=2045532 >
- Trajkovski, G. (2025, April 30). *Introduction to Adversarial AI* [Video]. Pluralsight. <https://app.pluralsight.com/library/courses/adversarial-ai-introduction/table-of-contents >
- Wylie, M. (2025, April 2). *The AI-Driven Cybersecurity Analyst* [Video]. LinkedIn Learning. <https://www.linkedin.com/learning/the-ai-driven-cybersecurity-analyst/threat-detection-with-ai?u=2045532 >
- Yawn, AJ. (2025, April 23). *Cybersecurity Foundations: Governance, Risk, and*

Compliance (GRC) [Video]. LinkedIn Learning.

<https://www.linkedin.com/learning/cybersecurity-foundations-governance-risk-and-compliance-grc-25657666/get-started-in-cyber-with-grc?resume=false&u=2045532 >

Student Course Page (WGU Portal)

Review the following helpful information and resources available on your course page.

- **Go to Course Materials:** This button will take you to the course materials where you will find the Tasks' requirements and instructional material on problem statements, project proposals, executive summaries, and creating a technical report.
- **Explore Cohort Offerings:** This button will allow the student to view the live cohort offerings and register for one or more cohorts that fit their schedule.
- **Instructor Information:** This section will allow the student to review the assigned course instructor information, office hours, email, and appointment scheduling link.
- **WGU Connect:** WGU Connect Communities is a platform that allows students, alumni, faculty, and staff of Western Governors University (WGU) to connect with each other, share resources, and engage in discussions related to their academic and professional interests. It provides a space for networking, collaboration, and support within the WGU community.

Opportunities to Learn

There are a variety of unique learning opportunities within this course. Use the following course features to build competence:

- Text presentation - explanations and instructions
- Images, infographics, and diagrams - visual tools to help students understand complex concepts
- Instructional videos
- Examples and case studies - realistic, relevant examples and case studies that brings content to life
- Activities - challenges to allow students to use what they have learned
- Lesson Quizzes - formative questions to give students practice and reinforcement of lesson-level concepts, with feedback

Course Schedule ***

Week	Course Chapter
1	1 – Course Overview 2 – Introduction to AI and Security
2	3 – Artificial Intelligence 4 – Machine Learning
3	5 – Understanding GenAI 6 - Neural Networks
4	7 – Data Management 8 - Cybersecurity
5	9 – AI and Cybersecurity 10 – Ethics in Artificial Intelligence
6	11 – Privacy for GenAI in Cybersecurity 12 – Accountability for GenAI in Cybersecurity
7	13 – The Future of GenAI in Cybersecurity 14 – Course Summary

***It is most important to know that many of the chapters in the online Zybooks text contain links and references to other reading materials, learning resources, graphics and videos that MUST be reviewed. It is important to review these materials as their content does provide a comprehensive presentation of the subject matter and is testable.